

CS PMOS Circuit Analysis

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Abstract— This report presents the design, analysis, and simulation of a common source (CS) PMOS amplifier, with emphasis on the circuit's DC and small-signal dynamics. The work begins with a symbolic analysis to determine the biasing behavior and key performance parameters. Based on transistor characteristics obtained from the GF180MCU process design kit, the required component values are calculated to meet the specified operating conditions. The circuit is then implemented and simulated in LTspice, and the calculated results are compared with simulation outcomes to assess the accuracy of the analytical model.

I. INTRODUCTION

This report follows a structured design and analysis approach for a common source (CS) PMOS amplifier. The work begins with deriving the symbolic relationships between the required circuit parameters through hand analysis. LTspice is then used to extract the main physical characteristics of the transistor models from the Global Foundries GF180MCU Integrated Circuit Process Design Kit. Using the specified current and voltage targets, the remaining circuit parameters are calculated analytically. The circuit is subsequently simulated using the calculated values, which are further adjusted to achieve the desired operating conditions. Finally, the calculated and simulated results are compared, and the observed differences are discussed.

II. SYMBOLICAL ANALYSIS

The symbolic analysis focuses on deriving expressions for the main quantities of the circuit shown in Fig. 1. These include the reference current (I_{REF}) through the resistor R , the bias current (I_{BIAS}) in the PMOS transistor, the small-signal gain (A_0), as well as the input resistance (r_{in}) seen at the PMOS gate and the output resistance (r_{out}) seen at the PMOS drain.

A. Reference current

The reference current is obtained from the NMOS drain current equation in saturation:

$$I_{REF} = \frac{1}{2} k'_n \frac{W_n}{L_n} (V_{GSn} - V_{Thn})^2 (1 + \lambda_n V_{DSn}) \quad (1)$$

The nodes G1 and S1 are shorted then $V_{GSn} = V_{DSn}$ so (1) would become:

$$I_{REF} = \frac{1}{2} k'_n \frac{W_n}{L_n} (V_{DSn} - V_{Thn})^2 (1 + \lambda_n V_{DSn}) \quad (2)$$

Where:

- k'_n : NMOS process transconductance parameter
- $\frac{W_n}{L_n}$: Transistor aspect ratio, defined as the width-to-length ratio of the NMOS devices
- V_{GSn} : Gate-to-source voltage of the NMOS transistor MN1
- V_{Thn} : Threshold voltage of the NMOS transistor
- λ_n : Channel-length coefficient of the NMOS transistor

- V_{DSn} : Drain-to-source voltage of MN1, which is equal to V_{GSn} due to the diode-connected configuration

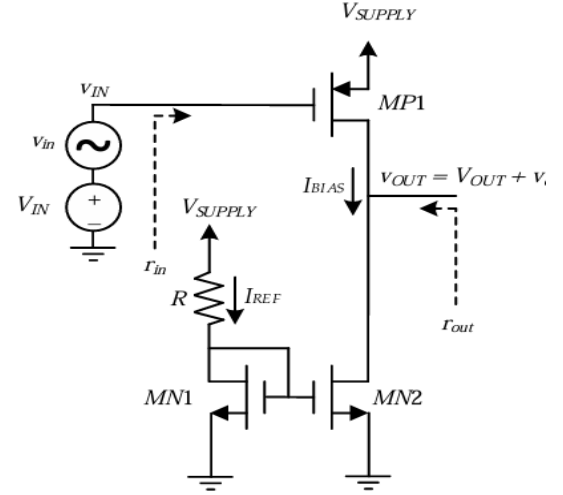


Fig 1. CS PMOS amplifier circuit

B. Bias current

Using the same approach, the following expressions are obtained for the MN2 transistor:

$$I_{BIAS} = \frac{1}{2} k'_n \frac{W_n}{L_n} (V_{GSn} - V_{Thn})^2 (1 + \lambda V_{out}) \quad (3)$$

C. Small signal gain

To determine the small-signal gain, the circuit of Fig. 2 is replaced by its small-signal equivalent model. In this model, the NMOS current source is represented as an ideal independent current source, which becomes an open circuit when disabled. In addition, the DC input voltage V_{in} and the supply voltage V_{supply} are set to zero and therefore modeled as short circuits.

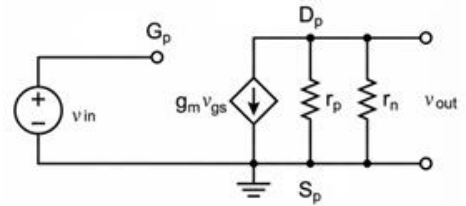


Fig 2. Small signal model

$$A_0 = -gm \times \frac{r_p * r_n}{r_p + r_n} \quad (4)$$

Where gm is the transconductance parameter of the PMOS, r_p and r_n are the resistances of the PMOS and the NMOS respectively.

$$gm = \frac{2 I_{BIAS}}{V_{GSn} - V_{Thp}} \quad (5)$$

$$r_p = \frac{1/\lambda_p + V_{SDp}}{I_{BIAS}} \quad (6)$$

$$r_n = \frac{1/\lambda_n + V_{out}}{I_{BIAS}} \quad (7)$$

D. Input resistance

The input resistance seen looking into the PMOS gate (Gp) in Fig. 2 is effectively an open circuit. Therefore,

$$r_{in} \approx \infty \quad (8)$$

E. Output resistance

To determine the output resistance, the input voltage v_{in} is set to zero, which in turn disables the dependent current source gmV_{gs} . Under this condition, the output resistance is given by the parallel combination of r_p and r_n :

$$r_{out} = \frac{r_p * r_n}{r_p + r_n} \quad (9)$$

III. TRANSISTOR PARAMETERS

To extract the NMOS and PMOS physical parameters, the methodology described in [1] is applied using the PDK. The supply voltage is set to 3.3 V, while both channel lengths are fixed at 0.28 μm . The transistor widths are initially chosen as 1 μm and are later modified to satisfy the required operating conditions.

A. NMOS parameters

Test circuit shown in Fig. 3

1) *Threshold voltage*: Using LTspice, the bias current I_{BIAS} is plotted while sweeping the gate-to-source voltage V_{GSn} , with the x-axis set to a logarithmic scale. The resulting plot is shown in Fig. 4. Based on this curve, a threshold voltage of $V_{Thn}=0.66\text{V}$.

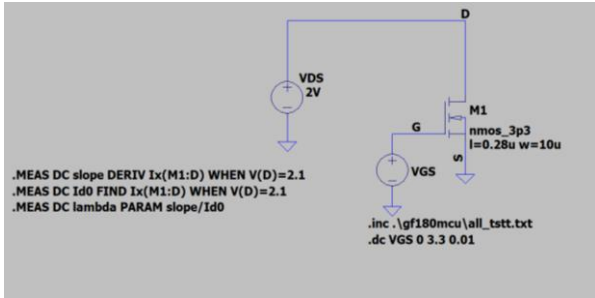


Fig. 3. NMOS test circuit



Fig. 4. I_{BIAS} with respect to V_{GSn}

2) *Length-width modulation*: To determine the channel-length modulation parameter λ_n , the gate-to-source voltage V_{GSn} is fixed near its expected DC operating point. Since this operating point is not initially known, a reasonable constant value is selected during the extraction process. Two drain-to-source voltage points and their

corresponding drain currents are then considered. The value of λ_n is obtained using the built-in LTspice measurement functions, following the standard channel-length modulation relation:

$$\lambda = \frac{i_{D2} - i_{D1}}{i_{D1}(V_{DS2} - V_{DS1})} \quad (10)$$

$$\lambda_n = 0.10919786 \text{ V}^{-1}$$

3) Device transconductance parameter

The transconductance parameter k is extracted using the NMOS saturation current equation while accounting for channel-length modulation. From the operating point simulation, the drain current I_D , gate-to-source voltage V_{GS} , and drain-to-source voltage V_{DS} are obtained. Using the extracted threshold voltage V_{Th} the overdrive voltage is defined as $V_{ov} = V_{GS} - V_{Th}$. The parameter k is then computed from the saturation current relation

$$k = \frac{I_D}{V_{ov}^2(1 + \lambda V_{DS})} \quad (11)$$

Substituting the operating point values obtained from LTspice yields a transconductance parameter of $k \approx 1.74 \times 10^{-3} \text{ A/V}^2$

4) Process transconductance parameter

The NMOS process transconductance parameter k'_n is obtained from the previously extracted device transconductance parameter k_n and the transistor geometry. The relationship between the two parameters is given by

$$k_n = \frac{1}{2} k'_n \frac{W_n}{L_n} \quad (12)$$

which can be rearranged as

$$k'_n = \frac{2k_n L_n}{W_n} \quad (13)$$

Using the transistor dimensions $W=10 \mu\text{m}$ and $L=0.28 \mu\text{m}$ along with the previously calculated value of k_n , the process transconductance parameter is found to be $k'_n \approx 9.73 \times 10^{-5} \text{ A/V}^2$

B. PMOS parameters

Test circuit shown in Fig. 5

1) *Threshold Voltage*: Using LTspice, the bias current I_{BIAS} is plotted while sweeping the PMOS gate-to-source control voltage V_{SGp} with the x-axis set to a logarithmic scale. The resulting curve is shown in Fig. 5. From this plot, the PMOS threshold voltage is extracted and a value of $V_{Thp} = 0.890\text{V}$.

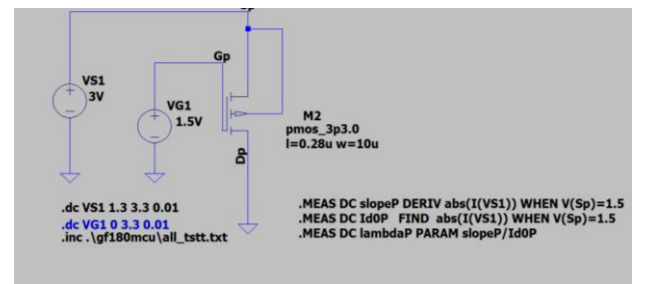


Fig. 4. PMOS test circuit

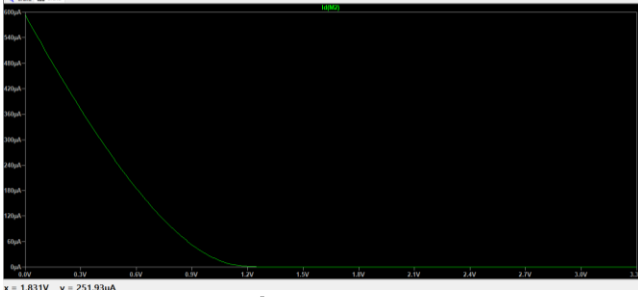


Fig. 5. I_{BIAS} with respect to V_{SGP}

2) *Length-width modulation*: To determine the channel-length modulation parameter λ_p the gate-to-source control voltage of the PMOS is fixed near its expected DC operating point. Since this operating point is not initially known, a reasonable constant value is selected during the extraction process. Two drain-to-source voltage points and their corresponding drain currents are then considered. The value of λ_p is obtained using the built-in LTspice measurement functions.

$$\lambda_p = 1.094V^{-1}$$

3) Device transconductance parameter

The PMOS transconductance parameter k_p is extracted using the PMOS saturation current equation while accounting for channel-length modulation. From the operating point simulation, the drain current I_D , source-to-gate voltage V_{SG} , and drain-to-source voltage V_{SD} are obtained. Using the extracted threshold voltage V_{thp} , the overdrive voltage is defined as $V_{ov} = V_{SG} - V_{thp}$. The parameter k_p is then computed from the saturation current relation

$$k_p = \frac{I_D}{V_{ov}^2(1 + \lambda V_{SD})}$$

$$k_p \approx 1.99 * 10^{-4} A/V^2$$

4) Process transconductance parameter

The PMOS process transconductance parameter k'_p is obtained from the device transconductance parameter k_p using the relation

$$k'_p = \frac{2k_p L_p}{W_p}$$

Using the transistor dimensions $W=10\mu m$ and $L=0.28\mu m$ along with the previously calculated value of k_p , the process transconductance parameter is found to be

$$k'_p \approx 1.11 * 10^{-5} A/V^2$$

IV- DC DESIGN CALCULATION

In this part, the extracted transistor parameters are used to design the circuit such that the required DC specifications are satisfied. Based on the student ID, the design parameter is calculated as $X=278X$ which leads to the target values

$$I_{REF}=1.39\text{ mA}, I_{BIAS}=278\text{ }\mu\text{A}, V_{out}=1.6\text{ V}, V_{DD}=3.3\text{ V}$$

1) NMOS current mirror transistor widths

The NMOS current mirror is formed by transistors MN1 and MN2, where MN1 is diode-connected and sets the reference current I_{REF} . Since both transistors operate in saturation and have the same channel length, the current ratio is given by:

$$\frac{I_{BIAS}}{I_{REF}} = \frac{(W/L)_{MN2}}{(W/L)_{MN1}} \quad (14)$$

Thus, the width of MN2 is chosen to be one-fifth of the width of MN1. A practical choice would be

$$L_{MN1} = L_{MN2} = 0.28\mu m; W_{MN1} = 25\mu m; W_{MN2} = 5\mu m$$

2) Resistor calculation

The resistor R sets the reference current I_{REF} through the diode-connected transistor MN1.

$$I_{REF} = \frac{1}{2} k'_n \frac{W_n}{L_n} (V_{DSn} - V_{Thn})^2 (1 + \lambda_n V_{DSn})$$

The gate-source voltage of MN1 is approximately $V_{GS1} \approx 1.19V$

The resistor value is therefore

$$R = \frac{V_{DD} - V_{GS1}}{I_{REF}} = \frac{3.3 - 1.19}{1.39\text{ mA}} = 1.52\text{ k}\Omega$$

3) PMOS width and input voltage

From previous parts: $V_{Thp} \approx 0.85V$; $\lambda_p \approx 0.70V^{-1}$

Choosing $V_{SG} = 1.5V \rightarrow V_{ovp} = 0.65V$

The required aspect ratio is $\frac{W}{L} \approx 54.3$, but L is fixed then

$W_{MP1} \approx 15\mu m$. The DC input voltage is

$$V_{IN} = V_{DD} - V_{SG} = 1.8V$$

V- SMALL SIGNAL PARAMETER CALCULATION

1) Voltage Gain

The PMOS transistor operates in saturation and the circuit behaves as a common-source amplifier with an active NMOS current-source load. The small-signal voltage gain is given by $A_v = -g_{mp}(r_{op}/r_{on})$ (15)

Using the extracted device parameters and the operating point, the PMOS transconductance is found to be $g_{mp} \approx 0.855\text{ mS}$. The output resistances are $r_{op} \approx 5.16\text{ k}\Omega$ and $r_{on} \approx 33\text{ k}\Omega$ yielding $A_v \approx -3.8V/V$.

2) Input resistance

The input is connected to the gate of the PMOS transistor, which ideally draws zero DC and small-signal current. Therefore, the input resistance is very large and can be approximated as $r_{in} \approx \infty$.

3) Output resistance

To find the output resistance, the input signal is set to zero (AC ground). The output node then sees the parallel combination of the PMOS and NMOS output resistances. Thus, $r_{out} = r_{op}/r_{on} \approx 4.46\text{ k}\Omega$

VI- SMALL SIGNAL PARAMETER CALCULATION

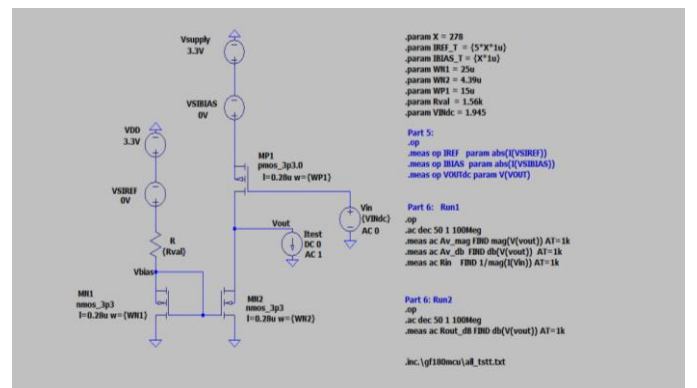


Fig. 6. CS PMOS circuit

1) Establishing the reference current

Once the bias loop was active, the reference current was set using the resistor RRR and the diode-connected transistor MN1MN_1MN1. The reference current is approximately: $I_{REF} \approx \frac{V_{DD}-V_{GS1}}{R}$

$$I_{REF\ target} \approx 1.39\text{mA}$$

Once the bias loop was active, the reference current was set using the resistor RRR and the diode-connected transistor MN1. The reference current is approximately: $R = 1.56\text{k}\Omega \rightarrow I_{REF} = 1.3998\text{mA}$ which closely satisfies the required specification.

2) Setting the bias current

The bias current is produced by mirroring I_{REF} through MN2M. With equal channel lengths, the ideal mirror ratio

$$\text{is: } \frac{I_{BIAS}}{I_{REF}} = \frac{(W/L)_{MN2}}{(W/L)_{MN1}} \approx \frac{W_{N2}}{W_{N1}}$$

Since practical effects such as channel-length modulation and the dependence on the drain voltage V_{out} slightly alter the mirror accuracy, W_{N2} was tuned iteratively while observing the DC operating point. The final value used was: $W_{N2} = 4.39\mu\text{m}$ which produced I_{BIAS} close to the required.

3) Adjusting the output voltage

The output voltage is set by the balance between the PMOS current source MP1 and the NMOS current sink MN2. The gate of MP1 is controlled by V_{in} , which determines V_{SG} and therefore the PMOS current. Increasing V_{in} reduces V_{SG} , weakening MP1 and pulling V_{out} downward, while decreasing V_{in} strengthens MP1 and increases V_{out} .

The target was $V_{out\ target} = 1.6\text{V}$

After tuning V_{in} , the final value obtained was:

$V_{in} = 1.945\text{V} \rightarrow V_{out} = 1.6045\text{V}$ which closely matches the desired DC output voltage.

4) DC operating point comparison

Quantity	Target Value	Simulated Value
I_{REF}	1.39 mA	1.3998 mA
I_{BIAS}	278 μA	274.6 μA
V_{out}	1.6 V	1.6045 V
V_{BIAS}	—	1.116 V

VII- AC SIMULATION RESULTS

1) Voltage gain A_v

The AC magnitude response of the amplifier was measured at 1 kHz with an input AC amplitude of 1 V. The simulated output magnitude was: 12.389472 dB which is equal to 4.17 in a linear scale. The phase response at

midband is approximately 180 degrees indicating an inverting amplifier. Therefore the simulated voltage gain is $A_v \approx -4.17\text{V}$

2) Input resistance

The input current magnitude was measured at 1 kHz as -194.06987 dB converting to a linear scale $|I_{in}| \approx 1.98 \times 10^{-10}\text{A}$. Since the input AC voltage is 1 V, the input resistance is $r_{in} \approx 5.05\text{G}\Omega \approx \infty$.

3) Output resistance

To measure the output resistance, a 1 A AC test current was injected at the output node while setting the input AC signal to zero. At 1 kHz, the measured output voltage magnitude was 74.361282 dB, converting to linear scale $r_{out} \approx 5.23\text{k}\Omega$

4) Comparison Table

Quantity	Calculated	Simulated
A_v	-3.8V/V	-4.17V/V
r_{in}	∞	5.05G Ω
r_{out}	4.46k Ω	5.23k Ω

VIII- DIFFERENCES BETWEEN CALCULATED AND SIMULATED

The differences between the calculated and simulated results are due to the simplifying assumptions used in hand analysis compared to the more accurate device modeling in simulation. The analytical model assumes constant small-signal parameters and ideal behavior, while the simulation includes bias-dependent effects, channel-length modulation, and body effect. As a result, the simulated gain and output resistance are slightly higher, and the input resistance remains extremely large, consistent with the high gate resistance of MOS transistors but limited by non-ideal effects in the model.

IX-REFERENCES

[1]: Jad G. Atallah, Mohammed Ismail, "Integrated Electronic Circuits," Springer, 2024.